

AI-assisted metadata curation and agent-ready OPeNDAP access to data

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1 Overview and Background

NASA Earth Science Data Systems (ESDS) manages 180+ petabytes of Earth science data today, with holdings expected to exceed 600 petabytes by the early 2030s. At that scale, operational problems become impossible to manage by manual review alone: metadata records, access guidance, and cloud-optimization artifacts can drift from the scientific data products they describe. The result is more failed discovery, more failed first attempts at access, and more staff time spent diagnosing collection-specific issues before researchers can begin useful scientific work.

ESDS already contains many of the pieces needed to address this problem. The Common Metadata Repository (CMR) provides authoritative collection and granule metadata; Hyrax and the OPeNDAP¹ Data Access Protocol (DAP) provide remote, subset-oriented access to the data themselves; Dataset Metadata Response ++ (DMR++) inventories expose the cloud retrieval structure [?]; and Distributed Active Archive Centers (DAACs) perform expert review and operational curation. What remains difficult is applying these capabilities consistently across large and growing archives, especially when the information needed to judge a collection is distributed across resources.

Artificial Intelligence (AI) is the right tool to address this problem because the task is not a single deterministic check; it is an evidence-gathering workflow across many heterogeneous collections. An AI agent can traverse collection structure, inspect CMR and OPeNDAP metadata, sample representative granules with OPeNDAP, compare findings across granules, draft machine-actionable guidance, and flag discrepancies for expert review. The factual claims in that workflow remain grounded in deterministic service responses and programmatic checks of metadata attributes at the variable level (e.g. units, coordinates) and cross-granule consistency.

The work described in Section 2 advances the ESDS AI Strategy primarily within the *Infrastructure* and *Access* pillars. It strengthens infrastructure by improving metadata systems, AI-readiness, and open machine-readable interfaces, and it strengthens access by making NASA data easier for both humans and AI agents to find and use through existing services. Instead of building a new portal or platform, this project modernizes the OPeNDAP Hyrax service and Earthdata Cloud (EDC) workflows.

1.1 Agentic inferencing

Before this call was released, we had already begun testing whether current foundation models could reason about remote data access and the metadata that makes an archive AI-ready, using OPeNDAP as the test case. We summarize that feasibility experiment here; the full audit trail is available in the open `grep-dap` repository [1].

The question was whether a Large Language Model (LLM), constrained by the OPeNDAP protocol and by direct statistical probes of the data, could recover missing semantic metadata the way a scientist would: reading directory and variable names, parsing data set structure, sampling values, and comparing against better-annotated sibling datasets. We ran the agent against a live Hyrax server at the University of Rhode Island (URI), grounding each conclusion in catalog structure, structural metadata, and small hypothesis-driven hyperslab probes. The results were strong:

1. On a collection conforming to the NetCDF Climate and Forecast Metadata Conventions (CF), the AI assistant audited 44 variables and flagged two *factually wrong* unit declarations—a

¹Open source Project for a Network Data Access Protocol (OPeNDAP) is the name of the 501(c)3 that maintains the Data Access Protocol (DAP) but the name OPeNDAP is often used to refer to that set of protocols.

wind speed declared `units="mm"` and a pixel count declared `units="C"`—by checking value ranges and sibling variables.

2. On an adjacent collection carrying *zero* semantic metadata, the AI assistant reconstructed units, coordinate orientation, and variable meaning by combining variable-name grammar, comparison with a related data set, and direct geographic probes.
3. For a derived gradient product, the AI assistant correctly estimated a physically essential quantity supplied nowhere in the metadata—the spatial support of the differencing operator—at 5 ± 2 km from three independent lines of evidence.

After being told only to learn the relevant OPeNDAP and Python Data Access Protocol (PyDAP) [2] interfaces, the model achieved the first two results from two open-ended audit prompts. Just as important, it separated *stated*, *inferred*, and *not-safely-guessable* attributes, graded each inference by evidence strength, and refused to invent unsupported information.

This subsection is included to demonstrate present-day *potential*, not to describe the operational method we propose for NASA. A prompt-driven audit can show feasibility, but it does not scale to agency needs. The approach in Section 2 therefore moves from this illustrative example to an autonomous agentic system that uses the same evidence discipline with explicit tooling, validation, and risk controls.

2 Proposed Approach

This project extends OPeNDAP/Hyrax in the EDC to improve discovery, metadata quality, and autonomous data access without replacing existing operational services. We will use model/API inference inside a tool-based agentic workflow that coordinates deterministic calls to CMR, OPeNDAP/Hyrax, and DMR++ documents². The design will keep model-specific logic behind stable interfaces.

In this architecture, the language model guides inspection, compares evidence, and drafts collection-level metadata documents. The factual claims are grounded in direct service responses, sampled data values, and programmatic checks of units, coordinates, ranges, fill values, chunk structure, and cross-granule consistency. Each generated claim will carry provenance, confidence, and review status. The software, schemas, workflow configuration, validation scripts, and sample artifacts will be made available as open-science outputs unless restricted by NASA policy.

The work is scoped to selected representative collections and granules chosen with our DAAC collaborator. The subsections below describe how the project first makes those collections crawlable by generic AI tools, then produces validated collection-level `usage.md` metadata documents. We originally intended to make these documents seamlessly available through the existing OPeNDAP service responses, but given time constraints of the project's funding,³ we will leave that integration as part of moving this work into operations. The generated `usage.md` documents will be available to external users through a separate S3 bucket.

2.1 AI-discoverable and crawlable service interfaces

The first part of the project will provide an AI-crawlable, web-friendly view of selected OPeNDAP-enabled ESDIS collections. The goal is for generic AI tools to discover target collections, traverse

²In previous work for Earth Science Data and Information System Project (ESDIS) we developed a service extension to Hyrax for DMR++ retrieval.

³EED3 contract funding ends on 29 January 2027.

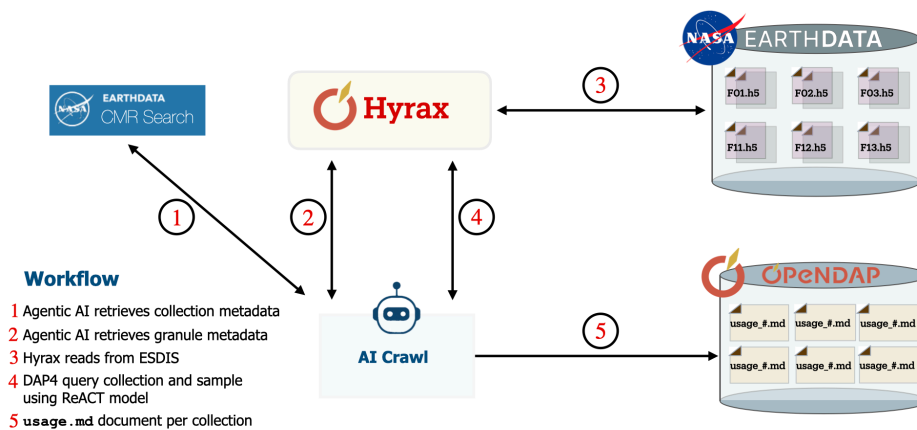


Figure 1: Deployment architecture for the agentic system. In part 2 of the project, the crawler/agent will run in AWS, The agent will query CMR and OPeNDAP/Hyrax and store the `usage.md` documents in AWS S3.

the collection hierarchy, and access data and metadata from a collection generic tools. This supports the *Infrastructure* and *Access* pillars by exposing existing services through open machine-readable interfaces rather than by building a new portal.

Hyrax already demonstrates this pattern when it serves local data files: its directory pages can be traversed by common web and AI tools. For the EDC environment, OPeNDAP has developed a Virtual Directory Interface that mimics the same Hyrax behavior for cloud-hosted collections. Early in the project we will work with ESDIS to deploy this interface.

2.2 AI-assisted metadata introspection and validation

The second part of the project will build and deploy an agentic workflow that inspects OPeNDAP-enabled EDC collections and drafts one collection-level `usage.md` file for each selected collection. We may crawl broader coverage when feasible, but the commitment of the project is to generate and validate guidance for a representative set selected with our DAAC collaborator.

Figure 1 illustrates the workflow. The agent will be configured with a schema for the `usage.md` documents and a set of tools to execute the various tasks required for metadata introspection and validation. The generated `usage.md` documents will be stored in an S3 bucket managed by OPeNDAP and separate from any of the DAAC artifacts.

For each selected collection, the workflow will query CMR, traverse Hyrax, inspect Data Access Protocol, Version 4 (DAP4) structural metadata, sample representative granules through OPeNDAP using tools such as PyDAP, inspect DMR++ access structure where available, run deterministic validators, compare evidence across granules, and draft collection `usage.md` documents. The resulting `usage.md` schema will include collection scope, provenance, source identifiers, variables, units, coordinates, fill values, valid ranges, temporal and spatial bounds, chunking and access guidance, known access issues, evidence links, confidence labels, review status, and last-refresh time.⁴

⁴For some collections, not all of this information will be relevant.

The agent may propose inferences, but every claim must be labeled as *stated*, *inferred*, *uncertain*, or *unsupported*. Unsupported claims will be omitted or routed for review rather than published. At no time with this system modify any EDC artifact (e.g., CMR record); only the `usage.md` documents will be created/edited by this system.

Safety and security are important concerns for this system and to guard against common security threats, the system will use a tool-based architecture with strict input/output schemas, and manual tool invocation [?] using, e.g., LangChain. This will provide a way to ensure that Web APIs surfaced by NASA systems (e.g., CMR) only receive sanitized inputs [?][?].

This part of the project also adds ‘staleness’ controls. Each `usage.md` document will include versioning, provenance, review status, and timestamps. While out of scope for this project, these features provide a path to future expansion to include support for Governance and Standards. We include them here because they simplify the review of the `usage.md` documents by a human expert.

Because DMR++ exposes chunk structure and related access information through the OPeNDAP interface, the guidance can also describe retrieval strategies that avoid unnecessary reads and align requests with chunked cloud objects where possible. That information is useful to autonomous agents and to virtual data systems such as VirtualiZarr [3].

The generated `usage.md` documents will be stored in an S3 bucket managed by OPeNDAP and separate from any of the DAAC artifacts. They will be available to external users through a separate S3 bucket, and they will be made available through the OPeNDAP/Hyrax service responses when the work is made operational.

3 Deliverables

During this project, we will produce three new concrete outputs:

1. **A working AI-crawable interface prototype.** We will deliver a Virtual Directory Interface for representative OPeNDAP-enabled ESDIS collections. The prototype will show that generic AI clients can discover target collections, traverse collection structure, and reach expected DAP4 service endpoints. If production deployment approval is outside the project’s control during the award period, we will deliver the working prototype and a documented EDC integration path.
2. **A collection of `usage.md` documents.** We will deliver a set of `usage.md` documents for selected collections in a separate S3 in the EDC bucket managed by OPeNDAP. The visibility of these documents will be controlled by OPeNDAP based on guidance by NASA ESDS.
3. **An open-source metadata-introspection repository.** We will deliver software that crawls selected collections, queries CMR, inspects OPeNDAP/Hyrax responses, samples representative granules, and drafts evidence-linked collection guidance. The public repository will include source code, schemas, prompts or workflow configuration, validation scripts, regression tests, developer documentation, and reproducibility notes.

Note: In the deliverables above we note that the deliverables will cover *selected* and/or *representative* collections and granules. This is to ensure the work and the promised deliverables are achievable within the project scope.

4 Validations and Path to Operations

Our validation approach follows from the three deliverables. For Part 1, we will validate the Virtual Directory Interface by confirming that generic AI clients can discover and traverse representative OPeNDAP-enabled collections in ESDIS without collection-specific adaptation. Success will be measured by the fraction of selected collections and granules that can be accessed.

For Part 2, we will validate the reachability of the key endpoints (i.e., CMR, OPeNDAP/Hyrax, and DMR++ documents) by the AI agents. In addition, we will validate the agentic metadata-generation workflow by comparing the generated markdown documents against authoritative sources, including CMR collection and granule metadata, and the results of direct DAP4 inspection. The system will be judged on whether it produces complete, evidence-linked metadata summaries, correctly flags uncertainty or disagreement, and avoids asserting facts that are not supported by the data or upstream metadata. The team has three expert validators that can review the generated documents.

Validation will also include presence of open-science products that are part of our public GitHub repository. These include scripts, test cases, generated sample guidance, and validation results. In addition, the `usage.md` documents themselves will be open-science artifacts that can be reviewed by the public and by domain experts. We will follow NASA policy/guidance on how publicly available these documents should be.

Deployment to operations is beyond the scope of this project. However, the path to operations is to embed the content of the `usage.md` documents into the standard responses provided by the OPeNDAP/Hyrax service. This will enable autonomous clients to use the embedded guidance to make correct data access and subsetting requests.

5 Timeline

This project is designed to be completed in 6 months and has two main parts. The work will start on August 3, 2026, and end on January 29, 2027, aligned with the EED3 contract funding period.

Part 1: AI-discoverable and crawlable service interfaces

Month 1 Deploy and test the candidate Hyrax virtual directory interface and define target validation collections.

Decision point: confirm that the interface is operational (as a working prototype).

Part 2: AI-assisted metadata introspection and validation

Month 2 Design the agent workflow, evidence schema, and 'usage.md' schema.

Month 3 Implement initial CMR, OPeNDAP/Hyrax, PyDAP, and DMR++ tool calls.

Month 4 Generate the first 'usage.md' artifacts for representative collections.

Month 5 Run expert review and revise validators based on the review.

Decision point: Confirm that the generated metadata documents are accurate and useful, and that the review process is effective.

Conclusion

Month 6 Package the open-source artifacts, final report, and presentation/demo, and present the working prototype as the final deliverable.

6 Team

The project team combines OPeNDAP engineering, NASA Physical Oceanography – Distributed Active Archive Center (PODAAC) operational knowledge, and physical-oceanography domain expertise. *James Gallagher* will lead the project and provide guidance on the Hyrax/OPeNDAP architecture, Virtual Directory interface, contribute to agentic AI development and software engineering. *Dan Holloway* will contribute agentic AI development and software engineering. Both Holloway and Gallagher have prior experience building AI systems. *Miguel Angel Jiménez-Urias* will collaborate on agentic AI development, use of PyDAPdomain review, and testing as an OPeNDAP-funded contributor. Miguel has a background in machine learning and physical oceanography.

Ed Armstrong at NASA Jet Propulsion Laboratory (JPL) PODAAC will support domain validation and testing, help select representative collections, and provide PODAAC operational feedback on whether the prototype fits practical DAAC review workflows. This gives the work a direct validation path against NASA ESDIS operational needs, rather than only an OPeNDAP-side software demonstration.

Peter Cornillon is the originator of the work described in the `grep-dap` project (Section 1.1) and will provide consultation as needed, as well as serving as a domain expert in Physical Oceanography, where he has published extensively.

Appendix A provides Table 1, which summarizes the team members, effort, and contributions.

7 Budget Summary

Note See Appendix B for Table 2 which summarizes the budget information.

Indirect costs For OPeNDAP, there are no indirect costs added to the budget since payment is through the existing ESDIS EED3⁵ contract. Ed Armstrong's cost does include indirect costs. See Appendix C for details.

Hardware procurement No hardware procurement is requested; existing infrastructure will be utilized.

Travel Travel is excluded from the budget.

Other direct costs Other direct costs include cloud compute resources for AI model and inference, estimated at \$10,000 over the project duration. This estimate is based on OpenAI model costs, LangChain usage and AWS cloud compute for testing.

Personnel Personnel costs are based on the estimated hours and hourly rates for the project team members, as detailed in the budget table in Appendix B. The total personnel cost is \$173,177. *James Gallagher* and *Dan Holloway* will be funded by the NASA EED3 contract. *Ed Armstrong* will be funded by the PODAAC. *Miguel Ángel Jiménez* will be funded by OPeNDAP as part of his role in OPeNDAP as the Scientific Community Director. *Peter Cornillon* will contribute unfunded domain validation and testing as a collaborator at OPeNDAP and the University of Rhode Island.

⁵NASA Earth Observing System Data and Information System Evolution and Development-3

A Team

Table 1: Team members and their contributions to the project.

Name	Effort	Role / Contribution
James Gallagher	2 months	Project lead; agentic AI development; software engineering
Dan Holloway	2 months	Agentic AI development; software engineering
Ed Armstrong	1 month	Domain expert and testing; key collaborator at PODAAC
Miguel Angel Jiménez-Urias	2 months ^a	Agentic AI development, domain expert and testing; collaborator at OPeNDAP
Peter Cornillon	Unfunded	Domain expert and testing; collaborator at OPeNDAP and the University of Rhode Island

^aFunded by OPeNDAP

B Budget table

Table 2: Summarized budget information for the project.

Who	Hourly rate	Months	Cost
James Gallagher	\$220.15	2	\$73,530
Dan Holloway	\$199.07	2	\$66,489
Ed Armstrong	See budget in Appendix C	1	\$33,158
	Monthly cost		
Compute	\$2,500.00	4	\$10,000
Tooling	n/a		n/a
Total			\$183,177

C Budget details for Ed Armstrong at JPL

AI-assisted metadata curation and agent-ready OPeNDAP access to data

COMPASS proposal

Timephased Cost Estimate Sheet

Dollars (Does not include Gov't Co-I's)

	Sep 2026 - May 2027		Total Program		
Hours / (FTEs)					
E. Armstrong (CO-I)	118	(0.09 FTE)	118	(0.09 FTE)	Hours / (FTEs)
Total Hours:	118	(0.09 FTE)	118	(0.09 FTE)	Subtotal
Amount	\$11,921		\$11,921		JPL Direct Labor Cost w/o Fringe
Fringe	\$6,348		\$6,348		Fringe
Category A	\$0		\$0		Cat A Direct Labor Cost
Total Direct Compensation (includes Employee Benefits)	\$18,269		\$18,269		Subtotal
Travel	\$0		\$0		Direct Travel Cost
JPL Services and Manufacturin	\$0		\$0		Direct Services Cost
Procurements					
Chargebacks	\$1,054		\$1,054		Direct Chargebacks cost
Subcontracts	\$0		\$0		Direct PS cost
Procurement RSA	\$0		\$0		Direct RSA cost
Purchase Orders	\$0		\$0		Direct PO cost
Caltech Transfers	\$0		\$0		Direct CT cost
Multi-Program Support	\$1,037		\$1,037		Direct MPS cost
Total Direct Costs	\$20,360		\$20,360		Subtotal
Allocated Direct Charge	\$9,340		\$9,340		Total ADC
General & Admin	\$3,458		\$3,458		Total G&A
Reserves (Burdened)	\$0		\$0		
Total JPL Costs	\$33,158		\$33,158		Subtotal
Government Co-I's Not in JPL's Costs	\$0		\$0		Bypass
Total Costs	\$33,158		\$33,158		Subtotal

References

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